Design & Simulation of a Smart College Network Using VLANs

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Abstract—This report presents the design and simulation of a segmented smart college campus network built on four VLANs (Students, Faculty, Admin, Guests). Inter-VLAN routing is provided by a Router-on-a-Stick configuration, access control is enforced with extended ACLs, and protocol behaviour is analysed with Cisco Packet Tracer Simulation Mode and Wireshark. All devices use static IP addressing; no DHCP service is deployed.

Index Terms—VLAN, Router-on-a-Stick, IEEE 802.1Q, ACL, Packet Tracer, Wireshark, campus network.

I. INTRODUCTION

A college campus serves users with very different trust levels and needs: students, faculty, administrative staff and visiting guests. Placing all of them on one flat network exposes sensitive resources, such as the college server and administrative machines, to every connected device. Virtual LANs (VLANs) solve this by dividing one physical switch into several isolated broadcast domains.

This project designs a small campus network around one core switch and one core router. Four VLANs are created, traffic between them is routed through a single trunk link using the Router-on-a-Stick method, and two extended Access Control Lists (ACLs) enforce who may reach what. The design is built and tested in Cisco Packet Tracer 8.x. Because Wireshark cannot capture Packet Tracer's internal simulated traffic, real protocol traffic (ARP, ICMP, DNS, TCP, UDP) is captured separately on a Linux host and analysed alongside Packet Tracer's Simulation Mode.

All hosts and the server use manually configured static IP addresses. No DHCP server is used anywhere in this project.

II. PROBLEM STATEMENT

A flat campus network places every user in a single broadcast domain. This creates four problems:

- **No isolation.** Students, faculty, administrative staff and guests share one segment, so any user can attempt to reach the college server or administrative machines.

- **Unnecessary broadcast traffic.** Broadcasts such as ARP reach every device, wasting bandwidth and exposing host information.
- **No access policy.** Without routing boundaries there is no point at which traffic between user groups can be filtered.
- **Poor manageability.** Group-specific rules cannot be applied or audited when all users are mixed together.

The problem addressed here is to design a segmented, policy-controlled campus network that separates user groups, still allows the permitted communication between them, and can be verified with test evidence.

III. OBJECTIVES

1. Design a star-topology campus network around one Cisco 2960-24TT switch and one ISR4331 router.
2. Create four VLANs (10 STUDENTS, 20 FACULTY, 30 ADMIN, 40 GUESTS) and assign switch access ports to them.
3. Provide inter-VLAN routing using Router-on-a-Stick with IEEE 802.1Q subinterfaces over a single trunk.
4. Assign static IP addresses to all hosts and the server, with each VLAN gateway on the router subinterface.
5. Enforce security with two inbound extended ACLs: STUDENT-ACL and GUEST-ACL.
6. Capture and analyse real ARP, ICMP, DNS, TCP and UDP traffic with Wireshark on a Linux host.
7. Test the complete network and document each result with screenshot evidence.

IV. SOFTWARE AND TOOLS

The simulation and the traffic analysis use the tools listed in Table I.

TABLE I. SOFTWARE, DEVICES AND TEST ENVIRONMENT

Component	Specification	Purpose
Simulation tool	Cisco Packet Tracer 8.x	Build, configure and test the campus network; Simulation Mode for ARP/ICMP packet flow
Core switch	College Core Switch, Cisco 2960-24TT	VLAN creation, access ports, trunk port Gig0/1

Component	Specification	Purpose
Core router	College Core Router, Cisco ISR4331	Router-on-a-Stick inter-VLAN routing and ACL enforcement
End devices	4 × PC-PT (Student, Faculty, Admin, Guest)	VLAN clients with static IP
Server	1 × Server-PT (College Server)	Protected resource in the ADMIN VLAN
Analysis tool	Wireshark 4.x	Real-traffic capture and protocol analysis
Host machine	HP Pavilion Laptop 16-af0xxx, Linux	Wireshark capture host
Capture interface	`wlo1` (Linux name for the wireless LAN adapter, equivalent to Wi-Fi on Windows)	Live packet capture
Host network	Mobile hotspot, `172.16.222.0/24`	Real-traffic test environment
Interface	Cisco IOS command line	Switch and router configuration and verification

V. NETWORK TOPOLOGY

The network uses a **star topology**: the College Core Switch (2960-24TT) is the central node, and every other device connects to it through its own dedicated link. The College Core Router connects to switch port Gig0/1, which is configured as an 802.1Q trunk. The four PCs and the server connect to access ports Fa0/1 to Fa0/5. A failure of one access cable affects only that device, and all inter-VLAN traffic passes through the single router link.

Fig. 1. Star topology of the smart college network in Cisco Packet Tracer (screenshot: topology view showing College Core Router, College Core Switch, Student PC, Faculty PC, Admin PC, Guest PC and College Server).

TABLE II. PHYSICAL CONNECTIONS

Device	Device port	Switch port	VLAN
Student PC	Fa0	Fa0/1	10
Faculty PC	Fa0	Fa0/2	20
Admin PC	Fa0	Fa0/3	30

Device	Device port	Switch port	VLAN
Guest PC	Fa0	Fa0/4	40
College Server	Fa0	Fa0/5	30
College Core Router	Gig0/0/0	Gig0/1	Trunk

The `show vlan brief` output confirms that Fa0/1 belongs to VLAN 10, Fa0/2 to VLAN 20, Fa0/3 and Fa0/5 to VLAN 30, and Fa0/4 to VLAN 40, with all remaining ports left in the default VLAN 1.

VI. VLAN AND IP ADDRESSING PLAN

Each VLAN has its own /24 network, with the gateway on the matching router subinterface (Table III). **No DHCP server is implemented.** Every PC and the server use static IP addressing, configured manually with IP address, subnet mask, default gateway and DNS server.

TABLE III. VLAN AND IP ADDRESSING PLAN

VLAN ID	Name	Network	Gateway (router subinterface)	Device, IP address, switch port
10	STUDENTS	192.168.10.0/24	192.168.10.1 (g0/0/0.10)	Student PC, 192.168.10.10, Fa0/1
20	FACULTY	192.168.20.0/24	192.168.20.1 (g0/0/0.20)	Faculty PC, 192.168.20.10, Fa0/2
30	ADMIN	192.168.30.0/24	192.168.30.1 (g0/0/0.30)	Admin PC, 192.168.30.10, Fa0/3; College Server, 192.168.30.100, Fa0/5
40	GUESTS	192.168.40.0/24	192.168.40.1 (g0/0/0.40)	Guest PC, 192.168.40.10, Fa0/4

Subnet mask for all networks: 255.255.255.0. DNS server: 8.8.8.8.

The `ipconfig` output on each PC (screenshots) confirms the static settings, for example the Student PC shows IPv4 address 192.168.10.10, mask 255.255.255.0 and default gateway 192.168.10.1.

VII. CONFIGURATION METHODOLOGY

The network was configured in four stages: VLANs on the switch, the trunk to the router, router subinterfaces for routing, and static addressing with inbound ACLs.

A. VLAN Creation and Access Ports

VLANs 10, 20, 30 and 40 were created and named, and each host port was set to access mode in its VLAN. Port assignments follow Table II.

```
vlan 10
 name STUDENTS
vlan 20
 name FACULTY
vlan 30
 name ADMIN
vlan 40
 name GUESTS

interface FastEthernet0/1
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/2
 switchport mode access
 switchport access vlan 20
interface FastEthernet0/3
 switchport mode access
 switchport access vlan 30
interface FastEthernet0/4
 switchport mode access
 switchport access vlan 40
interface FastEthernet0/5
 switchport mode access
 switchport access vlan 30
```

B. Trunk on Gig0/1

The switch port facing the router was set as an 802.1Q trunk so that VLANs 10, 20, 30 and 40 share one physical link.

```
interface GigabitEthernet0/1
 switchport mode trunk
```

C. Router-on-a-Stick Subinterfaces

The physical interface `g0/0/0` is brought up with no IP address. One subinterface per VLAN carries the `dot1Q` encapsulation tag and the VLAN gateway address.

```
interface g0/0/0
 no shutdown
interface g0/0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface g0/0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
interface g0/0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
```

D. Static IP Addressing

Each end device was configured manually in Packet Tracer (IP address, mask, gateway, DNS) using the values in Table III. DHCP was not configured on the router, switch or server.

E. Inbound ACLs

STUDENT-ACL and GUEST-ACL are applied **inbound** on the subinterfaces of the networks they filter, so traffic is checked as it enters the router from the source VLAN. The ACL contents are given in Section VIII.

```
interface g0/0/0.10
 ip access-group STUDENT-ACL in
interface g0/0/0.40
 ip access-group GUEST-ACL in
```

F. Verification Commands

```
show vlan brief
show ip interface brief
show access-lists
show ip interface g0/0/0.10 | include access list
show ip interface g0/0/0.40 | include access list
```

`show interfaces trunk` is a switch command and is not valid on the router; `show ip interface brief` was used on the router instead.

VIII. SECURITY POLICIES

Access between VLANs is controlled by the policy in Table IV, implemented as two named extended ACLs on the router.

TABLE IV. ACCESS POLICY

Source	Destination	Policy
Students (VLAN 10)	College Server, 192.168.30.100	Allow
Students (VLAN 10)	Rest of ADMIN network, 192.168.30.0/24	Deny
Students (VLAN 10)	All other destinations	Allow
Guests (VLAN 40)	Students, Faculty, Admin (192.168.10.0, 20.0, 30.0 /24)	Deny
Faculty (VLAN 20)	All	Allow (no ACL)

A. STUDENT-ACL (applied inbound on g0/0/0.10)

```
ip access-list extended STUDENT-ACL
 remark Allow Students to College Server only
 permit ip 192.168.10.0 0.0.0.255 host 192.168.30.100
 remark Deny Students to rest of Admin network
 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip 192.168.10.0 0.0.0.255 any
```

B. GUEST-ACL (applied inbound on g0/0/0.40)

```
ip access-list extended GUEST-ACL
 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip 192.168.40.0 0.0.0.255 any
```

C. Why Rule Order Matters

ACL entries are evaluated top to bottom and the first match is applied. In STUDENT-ACL the **host permit for 192.168.30.100 must come before the network deny for 192.168.30.0/24**. If the order were reversed, the deny would match first and Students would also lose access to the College Server, which sits inside the same /24. The final `permit ip ... any` is needed because every ACL ends with an implicit deny.

In GUEST-ACL the three deny rules cover all internal networks, and the final permit allows Guests to reach only destinations outside them.

D. Verification

The `show access-lists` output (screenshot) shows each rule with its match counter. After the tests, every entry of STUDENT-ACL (rules 10, 20, 30) and GUEST-ACL (rules 10, 20, 30, 40) showed 4 matches. `show ip interface g0/0/0.10` reports the inbound list as STUDENT-ACL and `g0/0/0.40` reports GUEST-ACL, with no outgoing list set on either.

IX. WIRESHARK ANALYSIS

A. Limitation and Approach

Wireshark cannot capture the internal simulated traffic of Cisco Packet Tracer, because those packets exist only inside the simulator and never reach a physical network interface. The analysis therefore uses two complementary sources: (1) **Packet Tracer Simulation Mode**, which shows the virtual ARP and ICMP packets between the simulated devices, and (2) **Wireshark on a real Linux host**, which captures live traffic on the physical interface `wlo1`. Both show the same protocols and the same request/reply behaviour.

In Simulation Mode, a PC in one VLAN first sends an ARP request for its own gateway (for example, who has 192.168.10.1), not for the final destination. To observe ARP followed by ICMP, the ARP cache was cleared with `arp -d` on the PC and `clear arp-cache` on the router, and the event list was filtered to ARP and ICMP only.

Fig. 2. Packet Tracer Simulation Mode event list showing ARP and ICMP events across the Student, Admin and Guest PCs, College Core Switch, Router and Server.

B. Real Capture Environment

TABLE V. WIRESHARK CAPTURE ENVIRONMENT

Item	Value
Host	HP Pavilion Laptop 16-af0xxx, Linux, user aryansjain18
Wireshark version	4.x
Capture interface	`wlo1` (Linux wireless LAN; equivalent to Wi-Fi on Windows)
Connection	Mobile hotspot, network 172.16.222.0/24
Host IP address	172.16.222.235/24 (shown by the ip a command for wlo1)
Hotspot gateway address	172.16.222.1 (hotspot gateway; see Section IX-E for the default route observed with ip r)
Capture file	`Day5_Wireshark_wlo1_172.16.pcapng`
Capture size	2968 packets

Android 12 and later hotspots assign addresses in the 172.16.x.x range rather than the older 192.168.43.x range, which explains the 172.16.222.0/24 network.

C. Capture Procedure

```
sudo apt install wireshark -y
sudo usermod -a -G wireshark $USER
```

After logging out and in again (or running `sudo wireshark`), interface `wlo1` was selected and the capture started. Traffic was generated with:

```
ping -c 4 172.16.222.1
ping -c 4 8.8.8.8
nslookup google.com
```

followed by browsing to google.com. The capture was then stopped and saved as `Day5_Wireshark_wlo1_172.16.pcapng`.

D. Display Filters and Observations

TABLE VI. DISPLAY FILTERS AND OBSERVATIONS

Filter	Protocol shown	Observation
`arp`	ARP	16 packets displayed (0.5%). Broadcast requests of the form "Who has 172.16.222.1? Tell 172.16.222.235" (42 bytes, Ethernet destination ff:ff:ff:ff:ff:ff, opcode request), and the hotspot device resolving 172.16.222.235 with replies carrying the laptop's MAC address.
`icmp`	ICMP	8 packets displayed (0.3%). Echo (ping) request, Type 8, from 172.16.222.235 to 8.8.8.8 (TTL 64) and Echo reply, Type 0, back (TTL 115), sequence 1 to 4, 98 bytes each. Ping to 8.8.8.8 succeeded with 0% loss, average round trip about 248 ms.
`dns`	DNS	94 packets displayed (3.2%). Standard queries for `www.google.com` (A, AAAA and HTTPS records) sent to the hotspot resolver and standard query responses returning Google IPv4 and IPv6 addresses. `nslookup google.com` resolved to 142.250.183.174.
`tcp`	TCP	Three-way handshake SYN, SYN-ACK, ACK to 142.250.x.x on port 443 when browsing google.com.
`udp`	UDP	UDP port 53 carrying the DNS queries and responses (for example source port 53 to an ephemeral destination port).

Fig. 3. Wireshark `arp` filter with packet details expanded (Ethernet II and Address Resolution Protocol layers).

Fig. 4. Wireshark `icmp` filter with Ethernet II, Internet Protocol Version 4 and ICMP layers expanded (echo request and reply to 8.8.8.8).

Fig. 5. Wireshark `dns` filter with packet details expanded (Ethernet II, IP, UDP and DNS response layers).

Fig. 6. Terminal output of `ip r | grep wlo1`, the ping to 8.8.8.8 and `nslookup google.com` on the capture host.

E. Observation on the Hotspot Gateway

The hotspot network is documented with gateway address 172.16.222.1. In the terminal capture, however, `ip r | grep wlo1` lists the default route via 172.16.222.42 (learned by DHCP from the phone's hotspot service), the `ping -c 4 172.16.222.1` returned

"Destination Host Unreachable", and the DNS answers in the capture come from 172.16.222.42. External connectivity through the hotspot still worked, as shown by the successful ping to 8.8.8.8. For this reason the 8.8.8.8 echo exchange, rather than a gateway ping, is used as the ICMP evidence. This DHCP lease applies only to the laptop's hotspot link; the simulated campus network itself uses static addressing only.

X. TEST CASES AND RESULTS

All six final tests behaved as designed: three communications were permitted and passed, and three access attempts were blocked by the ACLs as expected. Each test is backed by a Packet Tracer command-line screenshot, listed in the Evidence column.

Note on IP addressing: no DHCP is implemented. The test "Student PC receives a valid address" is therefore verified with `ipconfig`, which shows the manually configured static settings rather than a DHCP lease.

TABLE VII. FINAL NETWORK TEST RESULTS

Test	Command / check	Expected result	Actual result	Status	Evidence
Student PC receives a valid address	`ipconfig` on Student PC	Valid IP in 192.168.10.0/24, mask 255.255.255.0, gateway 192.168.10.1	Static: 192.168.10.10/24, gateway 192.168.10.1, DNS 8.8.8.8, manually configured	**Pass**	`27_Student_IPConfig_Static.png`
Student reaches its gateway	`ping 192.168.10.1` from Student	Reply	Reply from 192.168.10.1, bytes=32, TTL=255, 4/4 received	**Pass**	`28_Student_Gateway_Ping.png`
Permitted student-to-server communication	`ping 192.168.30.100` from Student	Pass: STUDENT-ACL rule 10 permits host 192.168.30.100	Reply from 192.168.30.100, TTL=127	**Pass**	`29_Student_Server_Allow.png`
Student attempts to access admin devices	`ping 192.168.30.10` from Student	Blocked: STUDENT-ACL rule 20 denies 192.168.30.0/24	Destination host unreachable from 192.168.10.1, 0/4 received. Hit count on `show access-lists` increases	**Blocked (expected)**	`30_Student_Admin_Blocked.png`

Test	Command / check	Expected result	Actual result	Status	Evidence
Guest attempts to access internal resources	`ping 192.168.10.10`, `192.168.20.10`, `192.168.30.100` from Guest	Blocked: GUEST-ACL denies 10, 20, 30	All three: destination host unreachable from 192.168.40.1, 0/4 received. Hits on GUEST-ACL rules 10, 20, 30	**Blocked (expected)**	`31_Guest_Internal_Blocked.png`
Faculty accesses permitted resources	`ping 192.168.30.100` from Faculty	Pass: no ACL on g0/0/0.20	Reply from 192.168.30.100, TTL=127, 4/4 received	**Pass**	`32_Faculty_Server_Allow.png`

Two details in the evidence are worth noting. On the first Student-to-Server ping, the first echo timed out and the remaining three were answered (3 of 4 received, TTL=127), which is the normal delay while the router resolves the server's MAC address by ARP. The TTL of 127 on inter-VLAN replies, against 255 for replies from the router itself, shows that the packet crossed the router (hop count reduced from 128).

A. Detailed ACL Test Matrix

Seven source-and-destination pairs were checked against the ACL rules.

TABLE VIII. ACL TEST MATRIX

#	Source	Destination	Expected	Actual	Reason
1	Student 192.168.10.10	Server 192.168.30.100	Allow	Reply	STUDENT-ACL rule 10 permit host
2	Student 192.168.10.10	Admin PC 192.168.30.10	Deny	Blocked	STUDENT-ACL rule 20 deny 192.168.30.0/24
3	Student 192.168.10.10	Faculty 192.168.20.10	Allow	Reply	STUDENT-ACL rule 30 permit any
4	Guest 192.168.40.10	Student 192.168.10.10	Deny	Blocked	GUEST-ACL rule 10
5	Guest 192.168.40.10	Faculty 192.168.20.10	Deny	Blocked	GUEST-ACL rule 20
6	Guest 192.168.40.10	Server 192.168.30.100	Deny	Blocked	GUEST-ACL rule 30
7	Faculty 192.168.20.10	Server 192.168.30.100	Allow	Reply	No ACL on g0/0/0.20

B. Connectivity Tests Before the ACLs

Before the ACLs were applied, every PC reached its own gateway, the Admin PC reached the server in the same VLAN (192.168.30.10 to 192.168.30.100), and inter-VLAN pings such as Student to Faculty, Student to Admin and Guest to Student all succeeded once Router-on-a-Stick was in place. The router's own pings to 192.168.10.10 and 192.168.30.10 returned 100% success (5/5).

C. Troubleshooting Log

TABLE IX. ISSUES ENCOUNTERED AND FIXES

Issue	Cause	Fix
Same-VLAN timeout between Admin PC and Server	Cables were on Fa0/6 and Fa0/8 instead of Fa0/3 and Fa0/5	Port labels enabled; devices rewired to match `show vlan brief`
Router interface flapping	Gig0/1 was not a trunk	Set `switchport mode trunk`
`show interfaces trunk` invalid on the router	Switch-only command	Used `show ip interface brief` on the router
Simulation Mode showed only ICMP	ARP entry already cached	`arp -d` on the PC, `clear arp-cache` on the router, ARP and ICMP filters
Simulation Mode showed only ARP	Simple PDU expired after the gateway ARP	Used `ping` from the PC inside Simulation Mode instead of a Simple PDU
Wireshark showed no Wi-Fi interface	Linux names the adapter `wlo1`, not Wi-Fi	Selected `wlo1`
Wireshark permission error	User not in the wireshark group	`usermod -a -G wireshark` and re-login, or `sudo wireshark`
Hotspot address 172.16.222.x instead of 192.168.43.x	Android 12+ hotspot range	Documented as an observation

XI. LIMITATIONS AND FUTURE IMPROVEMENTS

A. Limitations

- **No DHCP.** All hosts and the server use manually configured static IP addresses. This is simple and predictable for a small lab, but it does not scale and invites configuration errors.
- **No wireless access.** The design is wired only; no wireless LAN controller or access points are simulated, so wireless users and guest Wi-Fi are not represented.
- **Wireshark cannot see Packet Tracer traffic directly.** Bridging the simulated network to a real capture interface would need a Cloud device in Packet Tracer connected to a host adapter. Without it, real traffic on `wlo1` and simulated traffic in Simulation Mode were analysed separately.
- **Policy coverage.** ACLs are applied only on the Student and Guest subinterfaces. Faculty and Admin traffic is not filtered, so for example traffic from other VLANs toward the Guest network is not restricted.
- **Single points of failure.** One switch and one router link carry all traffic, with no redundancy.
- **Simulated environment.** Packet Tracer results confirm the logic of the design but do not measure real throughput or latency.

B. Future Improvements

TABLE X. PLANNED ENHANCEMENTS

Improvement	Benefit
DHCP service (router or server) with per-VLAN pools	Automatic addressing, fewer errors, easier scaling
Wireless LAN Controller and access points	Campus Wi-Fi mapped to the existing VLANs, including guest Wi-Fi
Additional IoT VLAN for campus devices (proposed, not implemented)	Isolates cameras, sensors and smart-building devices from user networks
IEEE 802.1X port-based authentication	Users and devices authenticate before receiving network access
Packet Tracer Cloud bridge to the host adapter	Allows Wireshark to capture the simulated network's traffic directly
Redundant links and a second switch	Removes single points of failure

XII. CONCLUSION

This project designed and simulated a smart college network that separates students, faculty, administration and guests into four VLANs (10, 20, 30 and 40), connected to one core switch in a star topology. A single 802.1Q trunk on Gig0/1 and four dot1Q subinterfaces on the ISR4331 provide inter-VLAN routing, and all devices use static IP addressing with no DHCP.

Two inbound extended ACLs enforce the security policy. STUDENT-ACL lets students reach the College Server at 192.168.30.100 while denying the rest of the ADMIN network, and depends on the host permit being placed before the network deny. GUEST-ACL stops guests from reaching the Student, Faculty and Admin networks. Screenshots of the final tests confirm three permitted communications (Pass) and three blocked attempts (Blocked, as expected), and the ACL match counters confirm the rules were hit.

Wireshark captures on the Linux interface `wlo1` (host 172.16.222.235/24 on the hotspot network) showed real ARP, ICMP, DNS, TCP and UDP behaviour, complementing Packet Tracer Simulation Mode, since Wireshark cannot capture Packet Tracer traffic directly. The work demonstrates that VLAN segmentation with Router-on-a-Stick and ACLs is an effective, low-cost way to isolate user groups, and it provides a base for DHCP, wireless, IoT segmentation and 802.1X in future work.